STA4001 2021 Spring midterm WANG Youmo.

1. Solution. $X_n=1$ if the team win and $X_n=0$ otherwise.

(a) Assume X_n is a Markov chain. we may find that

$0.8=P(X_{n+2}=1\,|\,X_{n+1}=1, X_n=1)=P(X_{n+2}=1\,|\,X_{n+1}=1)=P(X_{n+2}\,|\,X_{n+1}=1, X_n=0)=0.6$. Contradiction!

Therefore, we conclude that X_n is not a Markov chain. Define $Y_n=(X_n, X_{n+1})$ and the i.i.d. sequence $\{U_n, n=1,2,\cdots\}$ uniformly distributed on $[0,1]$. List the density of f as follows: $f((1,1),U)=\begin{cases}(1,1) & 0\le U\le 0.8\\ (0,1) & 0.8\le U\le 1\end{cases}$ $f((1,0),U)=\begin{cases}(0,1) & 0\le U<0.6\\ (0,0) & 0.6\le U\le 1\end{cases}$

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Thus, we can have $Y_{n+1}=f(Y_n, U_{n+1})$, where $\{Y_n, n=1,2,\cdots\}$ is a Markov chain.

(b) Let us list the state as $\{(0,0),(0,1),(1,0),(1,1)\}$. Then we have the transition matrix:

$$P=\begin{bmatrix}0.6 & 0.4 & 0 & 0\\ 0 & 0 & 0.4 & 0.6\\ 0.4 & 0.6 & 0 & 0\\ 0 & 0 & 0.2 & 0.8\end{bmatrix}$$

Using $\begin{cases}\pi P=\pi\\ \sum_{i=1}^{4}\pi_i=1\end{cases}$, we derive $\pi=\left[\frac{1}{6}\ \frac{1}{6}\ \frac{1}{6}\ \frac{1}{2}\right]$. Which is just the stationary distribution.

2. Solution. $D\sim$ Uniform$(0,20)$. Monday - Friday $c_p=20$ $c_v=10$ $c_s=5$ Order Quantity: Integer.

(a) By ~~coffe~~ corollary, $\frac{c_p-c_v}{c_p-c_s}=\frac{20-10}{20-5}=\frac{2}{3}$, the cdf of D is $F(x)=\frac{x}{20}$.

Let $F(x)=\frac{c_p-c_v}{c_p-c_s}$, we can have the optimal decimal order is $\frac{40}{3}$. As the order quantity should be integer. we need to compare the expected profit when the order is 13 and 14. Write ~~Cp profit~~ $c_p\cdot E[\min(D,q)]+c_s\cdot E[(q-D)^+]-q\cdot c_v$

When $q=13$, ~~profit $=20\times\frac{\frac{13\times12}{2}+13\times8}{20}+5\times\frac{\frac{13\times12}{2}}{20}-13\times10=$~~ (wrong way, continuous type)

$$\text{profit}=20\left[\int_0^{13}\frac{q}{20}\,dq+\int_{13}^{20}\frac{13}{20}\,dq\right]+5\left[\int_0^{13}\frac{13-q}{20}\,dq\right]-13\times10$$ (Use x instead, better off).

$$=\frac{169}{2}+(260-169)+\frac{169}{4}-\frac{1}{2}\times\frac{169}{4}-130=66.625.$$

Similarly, when $q=14$, profit $=66.5<66.625$. So the optimal order quantity is 13. The corresponding profit is 66.625.

(b) Let $P(13)$ be the single period profit when the order quantity is 13. The desired confidence interval is $[6662.5-1.96\sqrt{100}\,\sigma,\ 6662.5+1.96\sqrt{100}\,\sigma]$. where $\sigma^2=E[P^2(13)]-\{E[P(13)]\}^2$. There's no need to calculate it. (Integration process).

(c) If not order pineapple, the expected profit is: $20\left(\int_0^6\frac{x}{20}dx+\int_6^{20}\frac{6}{20}dx\right)+5\int_0^6\frac{6-x}{20}dx$

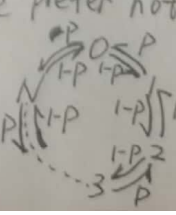
$=\frac{36}{40}\frac{36}{2}+6\times14+\frac{36}{4}-\frac{36}{8}=106.5$.

If order, despite the $c_f=40$ cost, $q=13$ is still the optimal ordering quantity. This time we just need to order another 7 items pineapple.

profit $=P(13)+13\times10-7\times10-40=86.625<106.5$. This means we prefer not ordering.

3. Solution Joe: 0 N friends right: p left: 1-p

(a) The transition diagram is a circle:



(b) From the symmetric property, all the states have the same stationary distribution. i.e., $\pi_0=\pi_1=\cdots=\pi_N$, for $\sum_{i=1}^{N}\pi_i=1$, we have $\pi_i=\frac{1}{N+1}$, $\forall i=0,1,\cdots N$.

(c) If $N+1$ is even, the DTMC is periodic, and the limit does not exist.
If $N+1$ is odd, the DTMC is aperiodic and the limit exists.

(d) $P=\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & \cdots \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & \cdots \\ \frac{1}{3} & 0 & 0 & \frac{2}{3} & 0 & \cdots \\ \frac{1}{4} & 0 & 0 & 0 & \frac{3}{4} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$ With $P_{i,1}=\frac{1}{i+1}$, $\forall i\in\mathbb{N}^+$. $P_{j,j+1}=\frac{j-1}{j}$, $\forall j\in\mathbb{N}^+\setminus\{1\}$. $P_{1,2}=1$. Then all else entries are 0.

(e) Yes, this DTMC is irreducible. For state k, $\{P^{k-1}\}_{1,k}=\prod_{j=2}^{k-1}\frac{j-1}{j}=\frac{1}{k-1}$. So 1 reaches k. Besides, $P_{k,1}=\frac{1}{k}$, so k reaches 1. 1, k communicate. Then we conclude all states are communicative so DTMC irreducible

(f) We've just proved that $\{X_n\}$ is irreducible. Define $f_{ij}^{n}=P\{T_j=n\,|\,X_0=i\}$. Notice there is only one path that firstly arrive at the state 1 given the initial state is state 1, so $f_{1,1}^{(n)}=P_{1,n}^{(n-1)}\cdot P_{n,1}=\frac{1}{n-1}\cdot\frac{1}{n}$. From $f_{1,1}^{(1)}=0$,

$f_1=\sum_{n=2}^{\infty}f_{1,1}^{(n)}=\sum_{n=2}^{\infty}\frac{1}{n-1}\cdot\frac{1}{n}=\sum_{n=2}^{\infty}(\frac{1}{n-1}-\frac{1}{n})=1$. So state 1 is recurrent. And

$E(T_1\,|\,X_0=1)=\sum_{n=2}^{\infty}n\cdot f_{1,1}^{(n)}=\sum_{n=2}^{\infty}\frac{1}{n-1}=\infty$. So state 1 is null recurrent.

4. Solution

(a) $P\{X_2=2\,|\,X_0=4\}=\sum_{i=1}^{7}P\{X_2=2\,|\,X_1=i\}P\{X_1=i\,|\,X_0=4\}=0.3\times0.2+0.2\times1+0.5\times0.2=0.36$.

(b) Positive recurrent states: $\{1,2,3\},\{7\}$; Null recurrent states: ϕ; Transient states: 4, 5, 6.

(c) We just need to look at the irreducible closed set $\{1,2,3\}$. $P=\begin{bmatrix}0 & 0.75 & 0.25\\ 0.8 & 0.2 & 0\\ 0 & 1 & 0\end{bmatrix}$.

Solve $\begin{cases}\pi=\pi P\\ \sum_{i=1}^{3}\pi_i=1\end{cases}$, we derive $\begin{cases}\pi_1=0.4\\ \pi_2=0.5\\ \pi_3=0.1\end{cases}$ Since restricted to $\{1,2,3\}$ It is aperiodic and positive recurrent.

The distribution of X_n converges to $(\pi_1,\pi_2,\pi_3)=(0.4,0.5,1)$.

(d) We need to see the probability finally arriving at a closed irreducible set. Let μ_i be the probability absorbed into state 7 that given initial state is i. Using the first step method

$R=\begin{cases}\mu_4=0.5\mu_5\\ \mu_5=0.4\mu_4+0.2\mu_6+0.2\\ \mu_6=0.5\mu_6+0.25\mu_5+0.25\end{cases}$ we derive $\begin{cases}\mu_4=\frac{3}{14}\\ \mu_5=\frac{3}{7}\\ \mu_6=\frac{5}{7}\end{cases}$ So the long run fraction of time for each state is $[\frac{4}{7}\times0.4,\frac{4}{7}\times0.5,\frac{4}{7}\times0.1,0,0,0,\frac{3}{7}]=[\frac{8}{35},\frac{2}{7},\frac{2}{35},0,0,0,\frac{3}{7}]$

And the long run average reward is $1\times\frac{8}{35}+2\times\frac{2}{7}+3\times\frac{2}{35}+7\times\frac{3}{7}=\frac{139}{35}$.

(e) Let t_i be the expected time arriving at any recurrent state given that the initial state is i. Using the first step method.

$\begin{cases}t_4=1+0.5t_5\\ t_5=1+0.4t_4+0.2t_6\\ t_6=1+0.5t_6+0.25t_5\end{cases}$ Then, $\begin{cases}t_4=\frac{16}{7}\\ t_5=\frac{18}{7}\\ t_6=\frac{23}{7}\end{cases}$

(f) Let N_i be the expected number of time steps that the Markov chain is in state 6, given that it starts in state i. Using the first step method.

$\begin{cases}N_4=0.5N_5\\ N_5=0.4N_4+0.2(1+N_6)\\ N_6=0.5(1+N_6)+0.25N_5\end{cases}$, it indicates $\begin{cases}N_4=\frac{2}{7}\\ N_5=\frac{4}{7}\\ N_6=\frac{9}{7}\end{cases}$.

5. Solution. Suppose we have two stationary distributions π and π'. Let $\alpha=\max\frac{\pi(j)}{\pi'(j)}$ and $j^*=\arg\min\frac{\pi(j)}{\pi'(j)}$. Then we derive $\pi(j^*)=\sum_i\pi(i)\{P^n\}_{i,j^*}=\sum_i\pi'(i)\frac{\pi(i)}{\pi'(i)}\{P^n\}_{i,j^*}$

The inequality turning into equality implies $\leq\sum_i\pi'(i)\alpha\{P^n\}_{i,j^*}=\alpha\pi'(j^*)=\pi(j^*)$

that $\{P^n\}_{i,j^*}(\alpha-\frac{\pi(i)}{\pi'(i)})=0$ for all n,i. Since it is irreducible DTMC, for any i we can find n such that $\{P^n\}_{i,j^*}>0$, leading to $\frac{\pi(i)}{\pi'(i)}=\alpha$. We must have $\alpha=1$ and hence, $\pi=\pi'$ consequently.